

Camp VeriFIRE: An Image and Weather Based Fire Risk Prediction System

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Abstract—Wildfire risk is growing worldwide, but convenient, localized risk information remains inaccessible to many outdoor enthusiasts. Because of this, I developed *Camp VeriFIRE*, an application that combines environmental images and local weather metadata to calculate a real-time fire risk score. Users upload a photo of their site, input location data automatically or manually, and the model integrates the temperature, humidity, and wind speed of the given region to issue a clear risk level. Deployed via a Flask web service with an interactive front end, Camp VeriFIRE provides campers with real-time situational awareness, potentially reducing the frequency of campfire-sourced wildfires.

Index Terms—Wildfires, Fire risk assessment, Convolutional neural networks, Computer vision, Weather Metadata, Environmental monitoring, Mobile applications

I. INTRODUCTION AND MOTIVATION

In recent years, California has witnessed an alarming surge in the frequency, intensity, and destructiveness of wildfires. The most relevant and recent of which, the Los Angeles fires, have displaced thousands, destroyed countless homes, and pushed emergency response systems to their limits. As climate change accelerates and urban development encroaches further into fire-prone areas, wildfires are no longer rare natural disasters but inevitable and constant occurrences.

This project, *Wildfire Risk Predictor*, was conceived in direct response to this growing crisis which has affected friends and family. Sparked by disruption caused by fires across Southern California, I set out to explore how machine learning and environmental sensing could offer a practical tool for early wildfire risk detection at the source.

My attention turned to anthropogenic fire sources which are significant contributors to the ignition of wildfires. In states like Idaho, for example, more than 50% of wildfires are caused by human activity, such as campfires and debris burning. Unattended or improperly managed campfires can ignite surrounding dry vegetation, which, when combined with wind and low humidity, can quickly escalate into uncontrolled wildfires. Such fires frequently occur in remote or rural areas, often near campsites, hiking trails, or sparsely inhabited private lands. Because these locations are isolated and lack robust infrastructure, fire detection and emergency response times tend to be slow. In some cases, fires can grow rapidly before local authorities are even alerted, especially if the fire starts during low activity periods or goes unreported. This delayed response, combined with dry vegetation and wind, can quickly turn a small, manageable fire into a widespread hazard. Recognizing

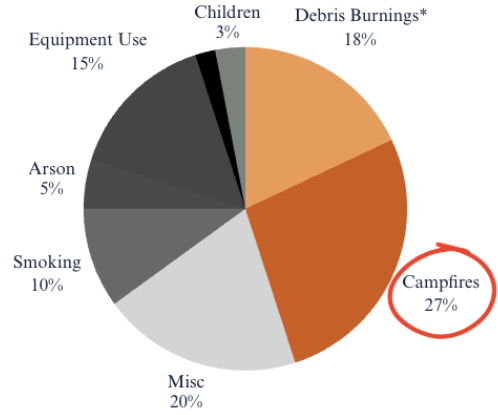


Fig. 1. "Common Causes of Wildfires." Idaho Firewise

this, I aimed to design my system with an understanding of how human tendencies intersect with geographic conditions. By accounting for these high-risk scenarios, especially where early detection and rapid response are difficult, my model can better reflect wildfire risk in practice.

By combining image-based vegetation analysis with localized environmental data such as wind speed, temperature, and humidity, the system aims to provide users with real-time assessments of wildfire risk in their immediate surroundings. The goal is to provide individuals, especially those in communities prone to fires, with predictive insight that can inform evacuation preparation, fire prevention, and awareness.

As wildfires become more frequent and unpredictable, tools like this can serve as an early line of defense, especially when deployed on mobile platforms for instant location-based feedback.

II. RELATED WORK

Deep learning models for wildfire detection have traditionally leveraged satellite imagery or combined visual and contextual data for event classification. However, these systems typically operate at large scales or require centralized deployments, limiting their accessibility for recreational users and site-specific risk assessment. Furthermore, many of these models provide limited interpretability, making it difficult for end-users to understand why a particular prediction was made.

Community-driven efforts such as Oguilmak's open-source wildfire prediction from satellite imagery demonstrate the po-

tential of machine learning pipelines built on publicly available datasets, but they remain tied to orbital platforms and do not provide fine-grained, site-level explanations. Similarly, industry and research groups have explored integrating weather and sensor data with image-based models. For example, Google Cloud reported on early efforts using TensorFlow to classify vegetation fuel conditions and combine them with meteorological data for predicting fire spread. These systems, while powerful, rely on specialized hardware and do not incorporate interpretable outputs for the user.

In contrast, Camp VeriFIRE shifts the paradigm to mobile, site-specific risk assessment using smartphone imagery combined with publicly accessible weather data. By integrating explainability techniques alongside the CNN predictions, the system allows users to understand which visual and environmental factors influence the fire risk score. This approach provides a more transparent, practical, and confidence-informed tool for individuals in fire-prone areas.

The USDA Forest Service’s 270-meter Wildfire Hazard Potential (WHP) map offers a national-scale index of wildfire hazard, derived from fire simulation models and historical data. While valuable for strategic planning, its coarse resolution and lack of interpretability make it less suitable for localized, real-time risk assessment. In contrast, Camp VeriFIRE provides a more granular and interpretable approach, enabling users to assess and understand wildfire risk at the individual site level.

III. METHODOLOGY

A. Data Collection and Preprocessing

A dataset of 300+ labeled outdoor images was compiled to represent fire-relevant environments. Labels reflected high fire risk (e.g., dry brush, coniferous forests) and low fire risk (e.g., moist rainforests, riversides). Images were sourced via Google Image Search using keyword queries generated by a large language model (LLM). Each image was resized to 128×128 pixels, normalized to $[0, 1]$, and augmented with flips, rotations, and brightness adjustments. Weather metadata (temperature, humidity, and wind speed) was obtained via the OpenWeatherMap API, using either user-provided city names or GPS coordinates.

B. Model Architecture

The image classifier uses MobileNetV2 as a convolutional backbone, initialized with ImageNet pretrained weights. On top of the frozen base network, a global average pooling layer was added, followed by a 30% dropout layer, a dense layer of 64 ReLU-activated units, and a final sigmoid neuron for binary fire risk classification. The network outputs a probability P_{image} representing the fire risk associated with the uploaded image.

C. Model Training and Implementation

The convolutional layers of MobileNetV2 were frozen during training to reduce overfitting on the relatively small dataset. The classifier head was trained using the Adam optimizer with binary cross-entropy loss, a batch size of 32, and 10 epochs. A 20% validation split was used, and performance metrics

included accuracy, precision, recall, F1-score, and AUC-ROC. Figure 2 shows the training and validation loss across epochs.

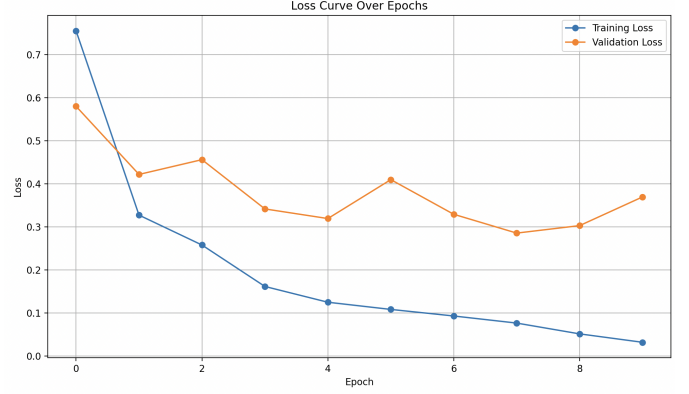


Fig. 2. Training and Validation Loss Across Epochs

D. Weather Integration and Risk Scoring

Alongside the image-based classification, I combine real-time weather conditions to adjust the fire risk score. I pull temperature, humidity, and wind speed data from the OpenWeatherMap API, using either the user’s city or GPS coordinates. These values are then used to scale the CNN’s probability output, with the idea that even a low-risk image can become hazardous under hot, dry, or windy conditions.

The way I handle this is fairly simple. I apply a set of multipliers to the image score when the weather conditions cross certain thresholds:

Here, T : temperature ($^{\circ}\text{C}$), W : wind speed (m/s), and H : relative humidity (%). The multipliers for each are defined as:

$$M_T = \begin{cases} 1.3 & \text{if } T > 32^{\circ}\text{C}, \\ 1.2 & \text{if } 21^{\circ}\text{C} < T \leq 32^{\circ}\text{C}, \\ 1 & \text{otherwise.} \end{cases}$$

$$M_W = \begin{cases} 1.5 & \text{if } W > 8 \text{ m/s}, \\ 1.2 & \text{if } 3.5 \text{ m/s} < W \leq 8 \text{ m/s}, \\ 1 & \text{otherwise.} \end{cases}$$

$$M_H = \begin{cases} 1.2 & \text{if } H < 30\%, \\ 1.1 & \text{if } 30\% \leq H < 40\%, \\ 1 & \text{otherwise.} \end{cases}$$

These thresholds are admittedly heuristic. I set them after some quick research into fire weather indices and patterns cited by agencies like the USFS and NPS, but they are not rigorously optimized or derived from historical fire datasets. The goal was to incorporate the most obvious weather drivers of fire spread without making the model overly complicated or data-hungry because I had not conducted formal research into the matter.

The final continuous risk score is calculated as:

$$\text{Risk Score} = P_{\text{image}} \times M_T \times M_W \times M_H$$

where P_{image} is the CNN’s predicted fire risk probability, and M_T, M_W, M_H are the temperature, wind, and humidity multipliers. This score is then bucketed into four user-facing categories: Low, Moderate, High, and Extreme

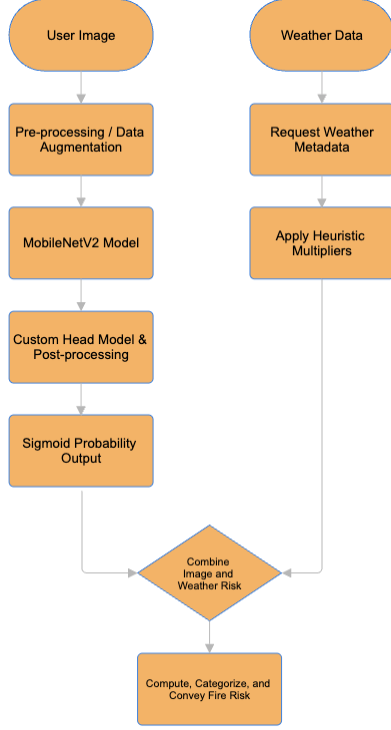


Fig. 3. High Level Project Design

E. Explainability of Predictions

To improve user trust and transparency, I integrated explainability techniques that highlight the evidence driving the model’s predictions. For the image classifier, we employed Gradient-weighted Class Activation Mapping (Grad-CAM) to produce heatmaps overlaid on the user-submitted photo. These visual explanations emphasize the vegetation or environmental features most influential in determining fire risk (e.g., dry brush, dense canopy). Figure 4 shows once such example.

By presenting users with both the categorical risk level and the visual evidence behind the decision, the system reduces the “black-box” nature of deep learning. Thus testers are able to understand both the prediction and rationale.



Fig. 4. Comparison of original site image (left) and Grad-CAM heatmap overlay highlighting regions contributing to the predicted wildfire risk (right)

F. Deployment

The system was deployed as a Flask web service exposing two endpoints: `/assess-risk` (image + city input) and `/verify-city`. The backend pipeline preprocesses user images, retrieves weather data, computes the adjusted risk score, and returns the category to the frontend. The web interface displays results along with relevant weather conditions.

Note: The current implementation is hosted locally and not deployed on a public or cloud server due to cost considerations. Future work includes deploying on scalable cloud infrastructure for broader access.

IV. RESULTS

Evaluation on a hold-out test set reveals approximately:

- Accuracy: 88.2%
- F1-Score: 0.86
- AUC-ROC: 0.91

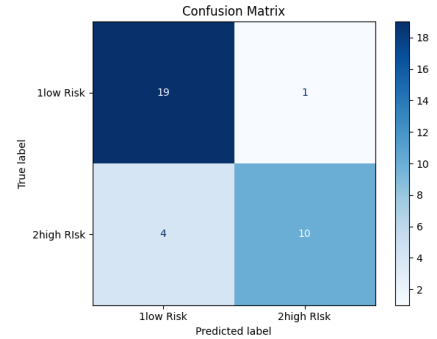


Fig. 5. Confusion matrix showing model performance on high-risk (class 1) and low-risk (class 0) predictions.

The confusion matrix shows the model is accurate overall (85%), with strong precision (90.9%) in predicting high risk cases. It correctly identifies most low risk cases but misses 4 high risk instances, suggesting it is cautious about flagging high risk, which may reduce false alarms but risks overlooking some critical cases. While this conservative approach reduces false alarms (a desirable trait for user adoption), false negatives can be quite dangerous. In wildfire prediction scenarios, false

negatives carry substantially greater consequences than false positives because an undetected high-risk situation could lead to catastrophic outcomes. Further training and testing can help alleviate this shortcoming by having a more robust model but also through threshold manipulation.

V. USER INTERFACE

As shown in Fig. 6, users upload a photo and optionally enable geolocation; the UI displays the computed risk level alongside weather details. The model runs in under 2 seconds per submit on the backend.

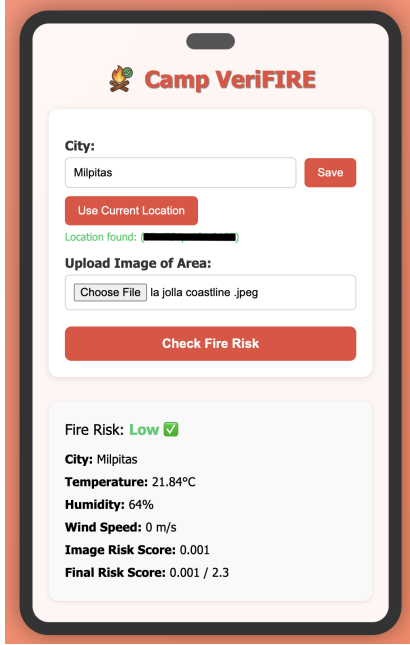


Fig. 6. Camp VeriFIRE User Interface.

VI. DISCUSSION

A. Design Choices

MobileNetV2 was selected as the backbone because it balances efficiency with strong performance on small datasets. To mitigate overfitting, its convolutional layers were frozen, and only a lightweight classification head was trained. The decision to integrate weather data alongside image predictions reflects the real-world role of environmental conditions in fire ignition and spread. Developing this system surfaced several key challenges: (1) sourcing labeled fire-risk imagery, (2) real-time fusion of visual and weather modalities, and (3) maintaining efficient inference in cloud environments. Despite these hurdles, the model achieved strong predictive performance while preserving user-centric interaction times.

B. Limitations

Several limitations affect the current system. First, the dataset of 300+ images is small relative to the variability of fire-prone environments, which constrains generalizability. Second, the integration of weather data relies on heuristic thresholds and multipliers. While these values were inspired

by fire weather guidelines from agencies such as the USFS and NPS, they were not optimized using historical wildfire datasets. As a result, the confidence of the weather-adjusted risk score is limited. Finally, the system currently outputs categorical risk levels without providing insight into which visual or environmental features contributed most to the decision.

C. Future Improvements

Future work will focus on several enhancements. First, the heuristic weather multipliers can be tuned systematically using grid search, Bayesian optimization, or calibration with historical fire incident data. This data-driven approach would more precisely reflect real-world fire risk patterns, reduce manual bias, and improve predictive robustness under varying environmental scenarios. Second, explainability techniques such as Grad-CAM or SHAP could be integrated to highlight image regions driving predictions, improving transparency and user trust. Third, comparisons with large-scale USDA wildfire risk maps (e.g., 270m $\times$ 270m resolution) would help validate how localized site-specific predictions differ from regional risk assessments.

Beyond these technical refinements, system-level extensions will also be explored. Mobile app support, crowd-sourced image datasets, SMS-based alerts, and the integration of additional data sources such as altitude and topographic features could further enhance the system's applicability. Partnerships with fire management agencies would provide robust validation and ensure that the tool aligns with real-world decision-making workflows. Ultimately, Camp VeriFIRE represents a step toward scalable, hybrid CNN-weather systems that deliver real-time, location-based wildfire risk alerts with both scientific rigor and practical utility.

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